Modelling multiple differences: a linear model with two or more categorical X variables

BIOL2022 - L04b

Dr Januar Harianto The University of Sydney

Modelling differences between groups: Part 2 (of 2)

Learning objectives

You should be able to:

- [ ] Frame ANOVA as a specific application of the General Linear Model (GLM) for categorical predictors.
- [ ] Formulate a GLM to test for differences between two or more group means.
- [ ] Explain how an ANOVA summary partitions the variance of a GLM with categorical predictors.
- [ ] Interpret main effects and interaction terms for categorical predictors within a GLM framework.
- [ ] Use post-hoc tests to probe significant main effects and interactions in a GLM.

Analysis of variance (ANOVA)

Using categorical predictors in a GLM

> It is better to solve the right problem approximately than to solve the *wrong* problem exactly.

— John Tukey (1915-2000)

The ANOVA model

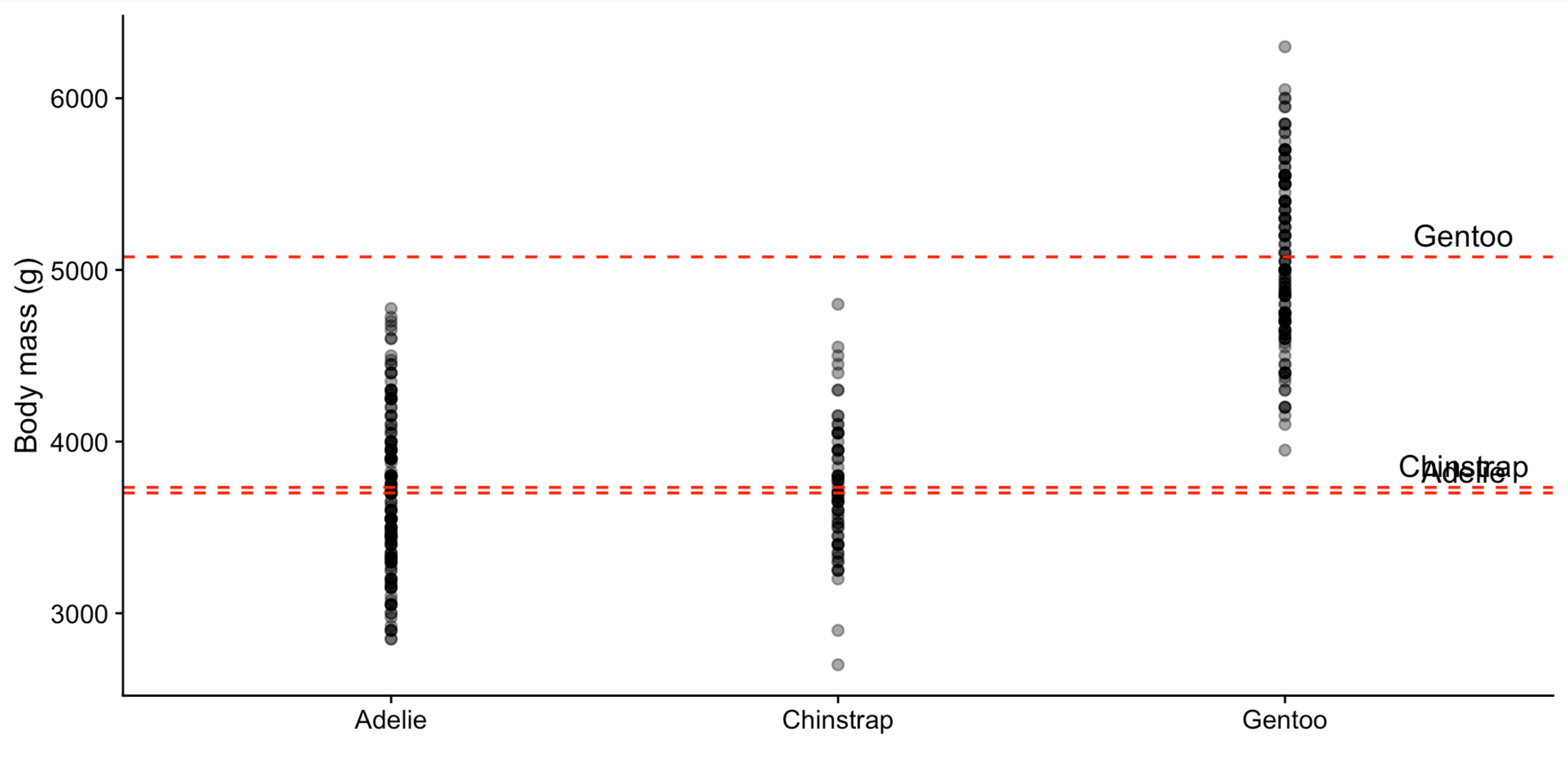
ANOVA is a type of linear regression that uses **categorical** predictors.

How it works (GLM perspective)

1. Fit a linear model using dummy variables for the categories.
2. Split the total variation in the response variable into:
   - Variation *between* groups (explained by the model).
   - Variation *within* groups (unexplained, or residual).
3. Compare these to test if the differences between groups are greater than the variation within groups.

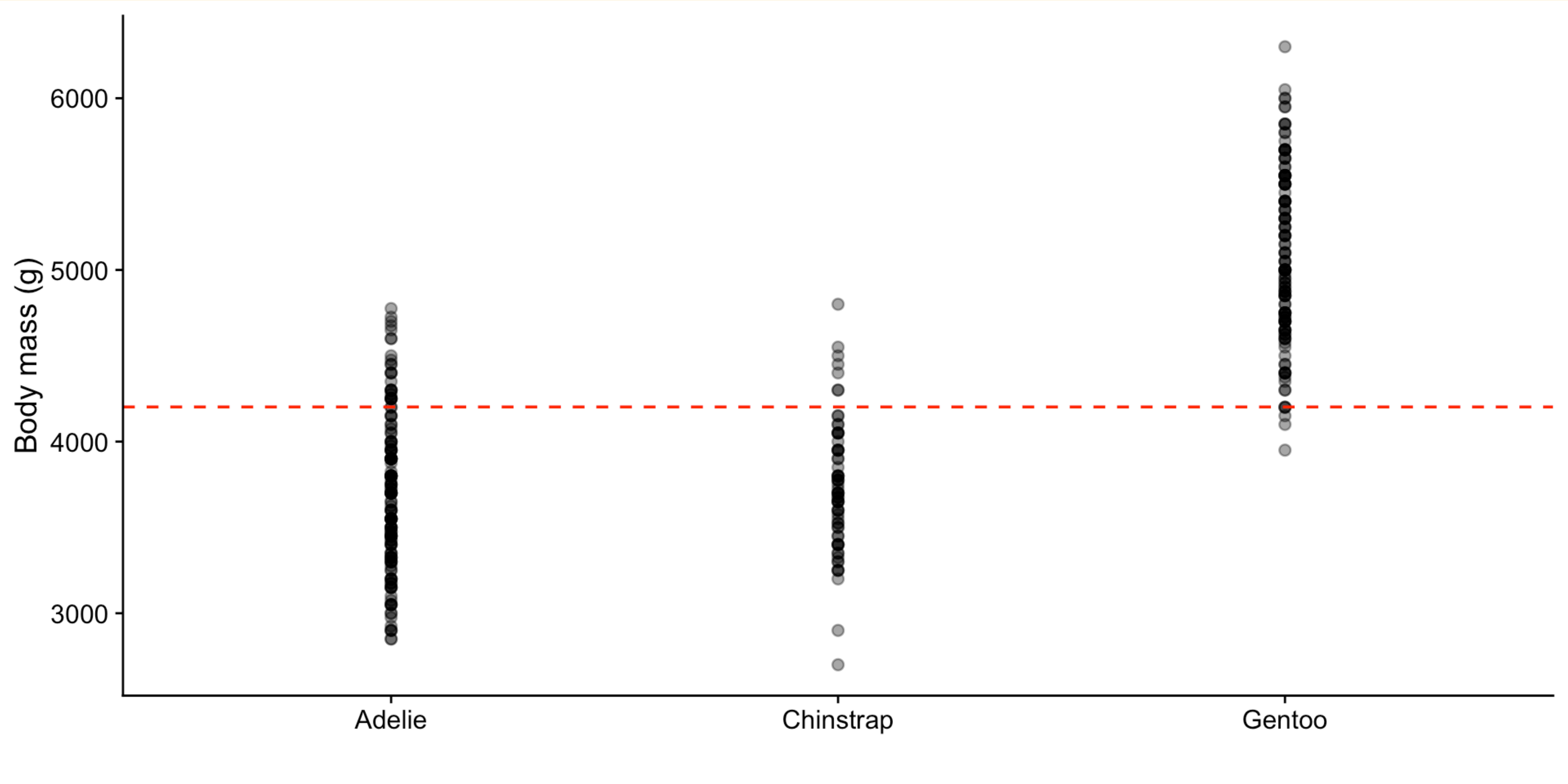
Variation between groups

Can we explain the variation in body mass by species (i.e. **are the means different**)?



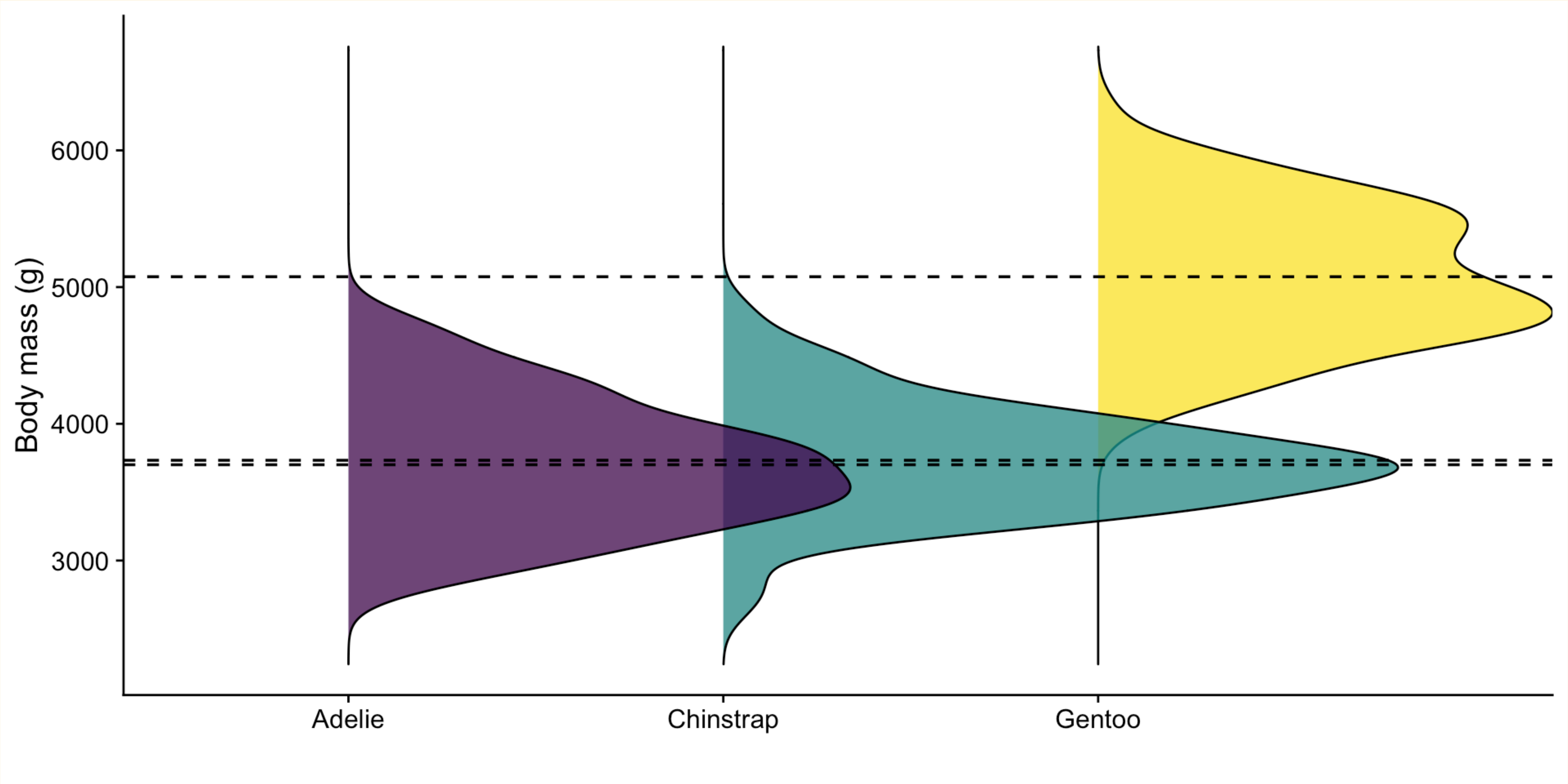
Variation within groups

Can we explain the variation in body mass by an overall mean (i.e. **are they all the same**)?



Another way to look at it

Are the distributions of body mass different enough to be considered separate?



Modelling the relationship

For a GLM:

$$\text{body mass} = \beta_0 + \beta_1 \cdot \text{species} + \epsilon$$

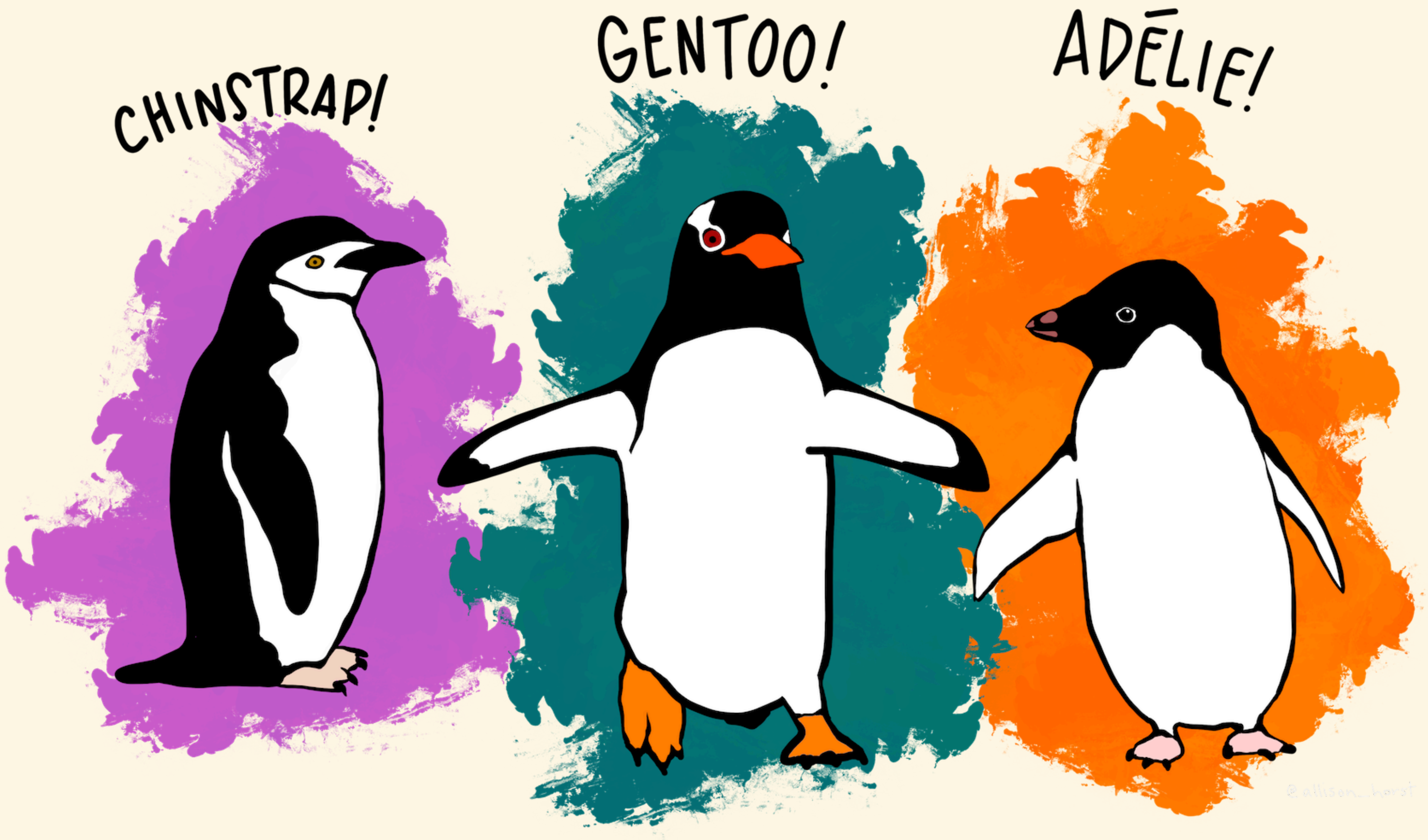
Once we use an ANOVA summary to process the model, we get:

$$\text{body mass} = \text{mean} + \text{species effect (i.e. difference)} + \epsilon$$

That is:

- β_0 is the **overall mean** of body mass.
- β_1 is the **difference** in body mass between species.
- We can add the mean to each species effect to get the **estimated body mass** for each species.

Example



Are species and sex significant predictors of body mass in penguins?

First, the general linear model

$$\text{body mass} \sim \text{species} + \text{sex}$$

The interaction term

Because we are interested in both predictor variables, we include an **interaction term**:

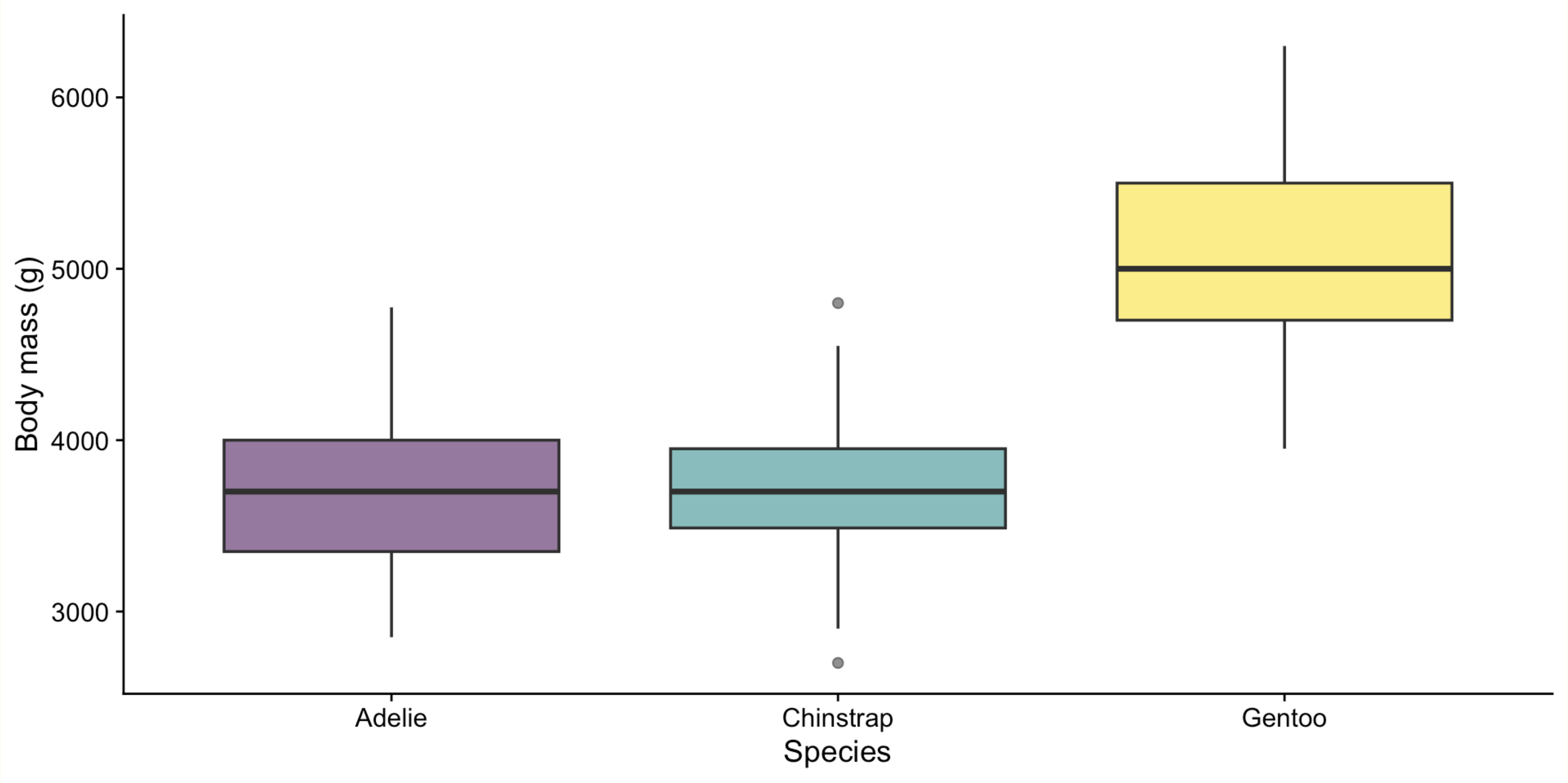
$$\text{body mass} \sim \text{species} \times \text{sex}$$

Four steps (as always), with a post-hoc

1. Visualise the relationship.
2. Fit the regression model.
3. Check and revise the model if needed (assumptions, outliers, issues).
4. Interpret the model ✨**and post-hoc**✨.

1 of 4: Visualise the relationship

$$\text{body mass} \sim \text{species} \times \text{sex}$$



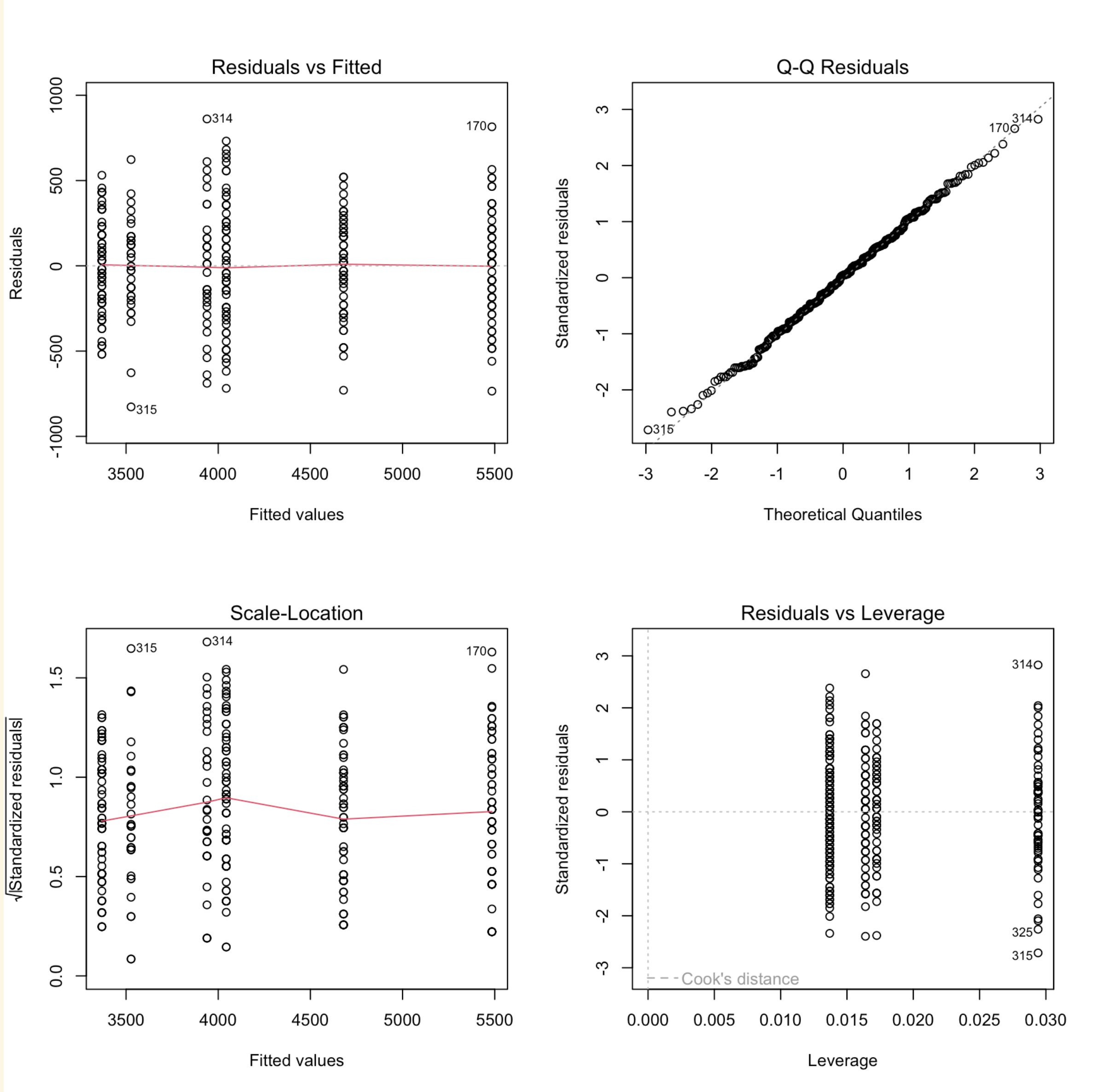
Side note: Beyond bar and box plots

2 of 4: Fit the model

Ths is performed using software so we can assume this is done correctly (for now). The model is still:

$$\text{body mass} \sim \text{species} \times \text{sex}$$

3 of 4: Check assumptions



Diagnostic plots for the linear model with species and sex as predictors of body mass.

4: Interpret

In practice, if we are interested in comparing means, the **ANOVA summary** table is the most useful.

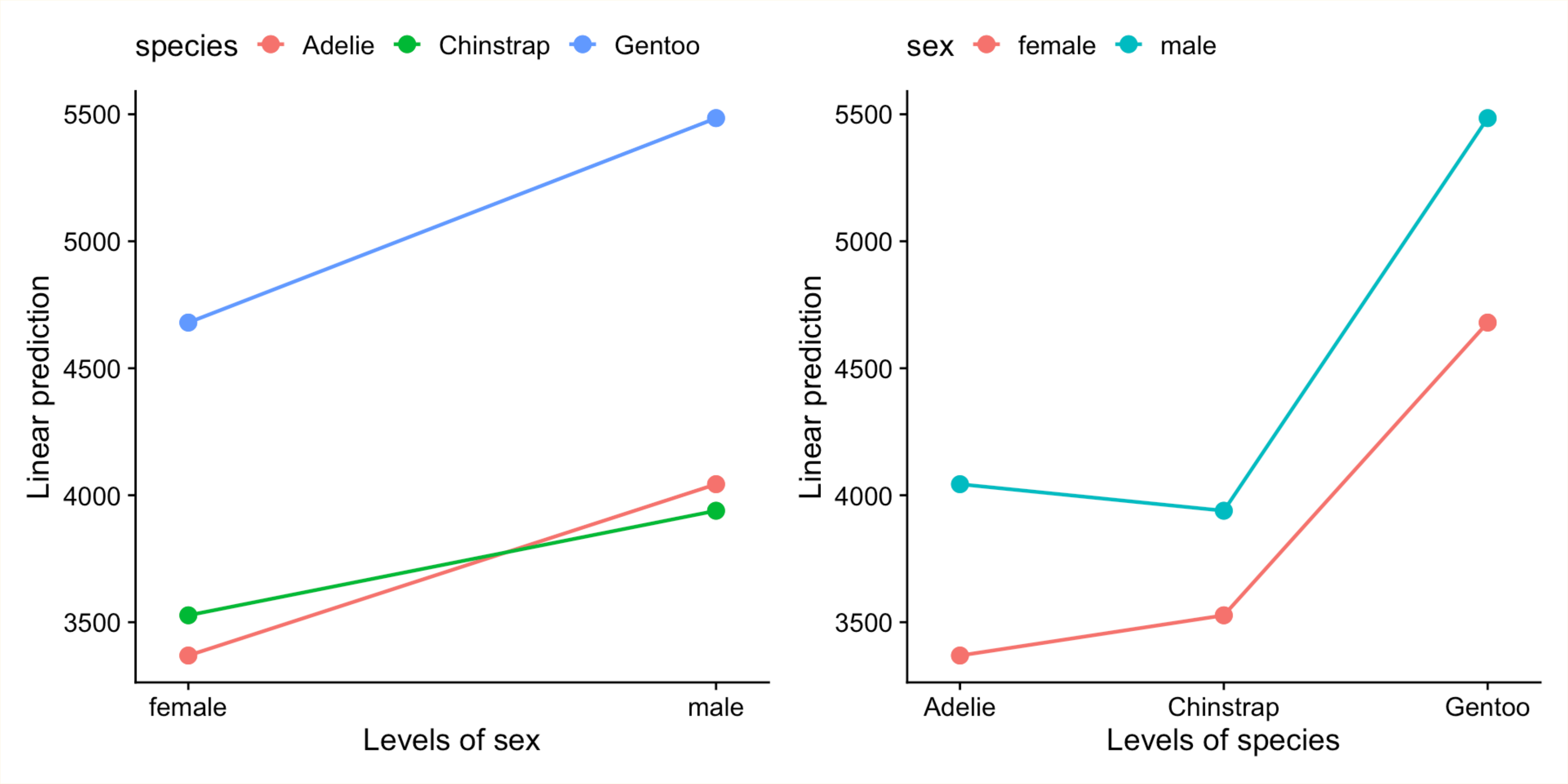
```
Analysis of Variance Table

Response: body_mass_g
             Df    Sum Sq  Mean Sq F value    Pr(>F)    
species       2 145190219 72595110 758.358 < 2.2e-16 ***
sex           1  37090262 37090262 387.460 < 2.2e-16 ***
species:sex   2   1676557   838278   8.757 0.0001973 ***
Residuals   327  31302628    95727                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There are **significant interactions**, so we no longer interpret the main effects. **Interpret the interaction term instead.**

Interpreting interactions

We want to check two things:

- Whether the relationship between body mass and species is dependent on **sex**, and/or
- Whether the relationship between body mass and sex is dependent on **species**.



Interpreting interactions

- Species affects the relationship between body mass and sex. Males are generally heavier than females, but the difference is less pronounced in Chinstrap penguins as seen by the flatter slope compared to the other two species.
- Sex also affects the relationship between body mass and species. Adelie males are generally heavier than Chinstrap males, but in females it is the opposite: Adelie females are generally lighter than Chinstrap females.

But which interactions are “significant”?

What if we look at the non-ANOVA summary?

```
Call:
lm(formula = body_mass_g ~ species * sex, data = penguins)

Residuals:
    Min      1Q  Median      3Q     Max 
-827.21 -213.97   11.03  206.51  861.03 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)               3368.84      36.21  93.030  < 2e-16 ***
speciesChinstrap           158.37      64.24   2.465  0.01420 *  
speciesGentoo             1310.91      54.42  24.088  < 2e-16 ***
sexmale                    674.66      51.21  13.174  < 2e-16 ***
speciesChinstrap:sexmale  -262.89      90.85  -2.894  0.00406 ** 
speciesGentoo:sexmale      130.44      76.44   1.706  0.08886 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Post-hoc

Where are the differences?

We are looking at the species × sex interaction where the relationship between species and body mass differs by sex.

Female penguins

Which species differ?

Male penguins

Which species differ?

Post-hoc comparisons identify the differences, their direction, and their size.

Pairwise comparisons within each sex

Estimated marginal means are the group means predicted by the fitted GLM.

Female penguins

- Adelie vs Chinstrap
- Adelie vs Gentoo
- Chinstrap vs Gentoo

Male penguins

- Adelie vs Chinstrap
- Adelie vs Gentoo
- Chinstrap vs Gentoo

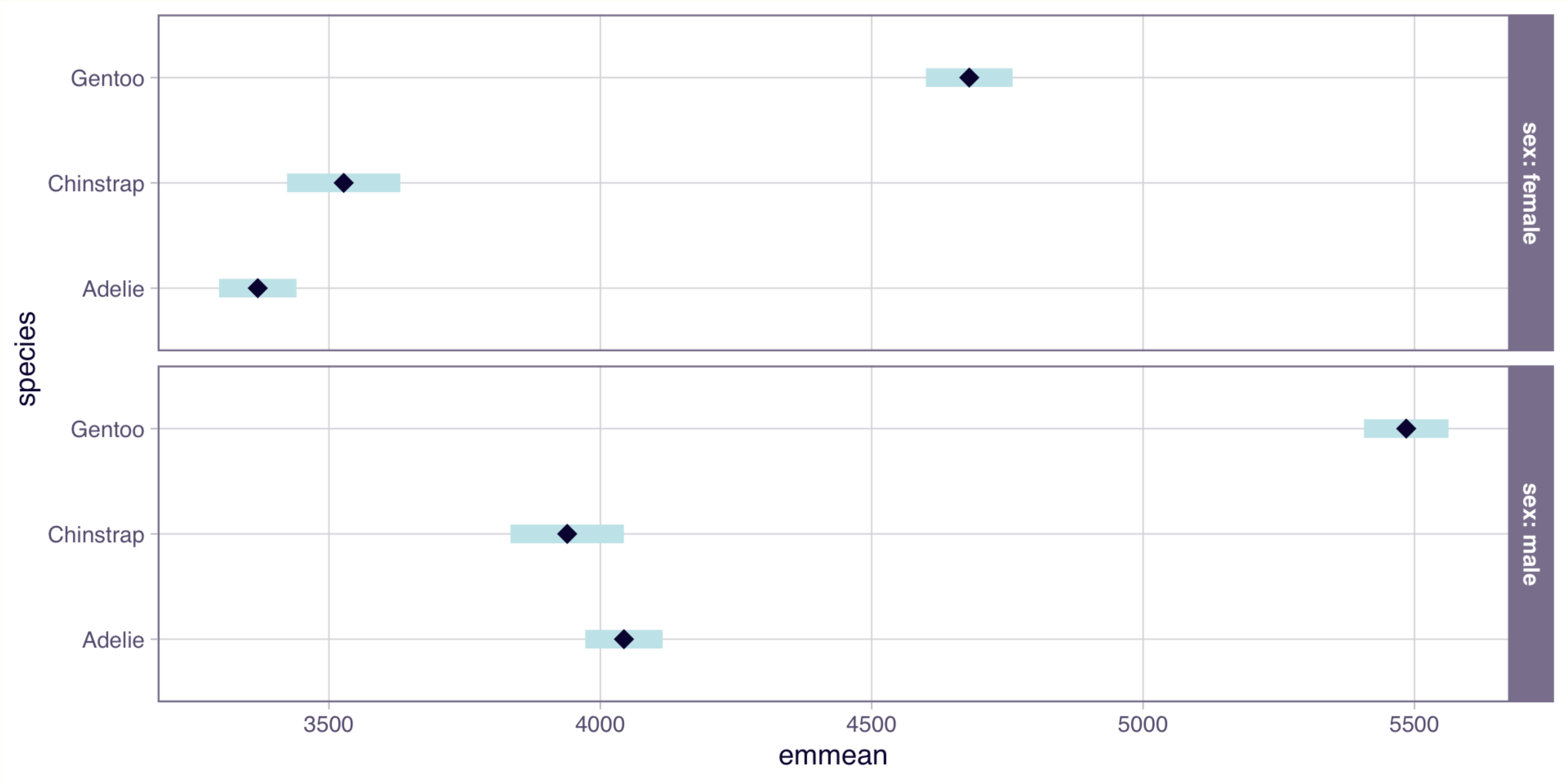
We can also consider pairwise comparisons within each species.

Results

```
sex = female:
 contrast            estimate   SE  df lower.CL upper.CL t.ratio p.value
 Adelie - Chinstrap      -158 64.2 327   -309.6    -7.12  -2.465  0.0377
 Adelie - Gentoo        -1311 54.4 327  -1439.0 -1182.77 -24.088 <0.0001
 Chinstrap - Gentoo     -1153 66.8 327  -1309.9  -995.19 -17.246 <0.0001

sex = male:
 contrast            estimate   SE  df lower.CL upper.CL t.ratio p.value
 Adelie - Chinstrap       105 64.2 327    -46.7   255.77   1.627  0.2357
 Adelie - Gentoo        -1441 53.7 327  -1567.7 -1314.98 -26.855 <0.0001
 Chinstrap - Gentoo     -1546 66.2 327  -1701.8 -1389.96 -23.345 <0.0001

Confidence level used: 0.95
Conf-level adjustment: tukey method for comparing a family of 3 estimates
P value adjustment: tukey method for comparing a family of 3 estimates
```

Are there **differences** between species significant for each sex, and if so, how did interactions affect the **direction** and **size** of the differences?

Results



Gentoo
Chinstrap
Adelie
Gentoo
Chinstrap
Adelie
species
sex: female
sex: male
3500
4000
4500
5000
5500
emmean

Questions to consider

- When is it more appropriate to summarise a GLM with an ANOVA table versus regression coefficients?
- How do the standard GLM assumptions (LINE) apply when working with categorical predictors?
- How does the GLM formulation and interpretation change when including one vs. two categorical predictors, especially regarding main effects and interactions?
- What does a significant interaction term in a GLM tell us about the relationship between categorical predictors?
- Why are post-hoc tests necessary for interpreting GLMs with significant categorical predictors with more than two levels?

Thanks!